

TUGUEGARAO, Philippines

WMO: 982330

Lat: 17.6478N

Lon: 121.7585E

Elev: 21

StdP: 101.07

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.8	19.7	16.4	11.7	19.8	17.3	12.4	20.5	6.9	26.5	5.9	24.5	1.0	0	0.272

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	9.5	36.9	27.9	36.0	27.7	35.1	27.5	28.7	35.4	28.2	34.7	27.9	34.0	1.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.9	22.6	33.4	26.5	22.0	32.7	26.1	21.5	32.1	93.3	35.5	91.0	35.0	89.3	34.3	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.2	5.0	4.0	DB	17.0	38.3	1.5	1.0	15.9	39.0	15.0	39.5	14.2	40.1	13.1	40.8	
			WB	15.7	29.7	1.4	1.0	14.7	30.4	13.9	31.0	13.1	31.5	12.2	32.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.5	24.2	25.2	27.0	29.3	29.9	30.0	28.9	28.7	28.4	27.2	26.2	24.6
	DBStd	2.53	1.86	2.01	1.96	1.80	1.60	1.19	1.37	1.33	1.25	1.46	1.57	1.61
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6383	440	427	528	579	618	601	586	580	552	533	486	453
	CDD18.3	3341	182	193	269	329	360	351	327	322	302	274	236	195
	CDH23.3	32708	987	1384	2451	3789	4240	4288	3641	3522	3167	2428	1791	1021
	CDH26.7	13349	218	468	1028	1848	2092	2106	1535	1405	1193	771	491	196
Wind	WSAvg	1.2	1.1	1.1	1.2	1.2	1.2	1.2	1.1	1.1	1.1	1.3	1.1	1.1
Precipitation	PrecAvg	1728	39	27	28	46	127	158	213	230	207	272	259	120
	PrecMax	2694	205	184	135	239	408	417	475	532	470	657	959	452
	PrecMin	1154	4	0	0	0	8	27	47	55	43	16	12	0
	PrecStd	468	44	38	31	53	80	79	107	131	103	169	236	112
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.8	33.6	35.5	37.9	38.3	37.6	36.4	35.4	34.5	34.0	32.9	31.5
		MCWB	26.2	26.8	27.6	27.5	27.8	28.7	28.3	27.8	27.6	27.2	27.0	26.0
	2%	DB	30.2	32.4	34.4	36.6	37.1	36.3	35.1	34.2	33.7	32.7	31.8	30.0
		MCWB	25.4	26.2	26.8	27.3	27.9	28.2	27.8	27.5	27.2	26.8	26.5	25.6
	5%	DB	29.0	31.1	33.3	35.5	36.0	35.4	34.0	33.4	33.0	31.7	30.7	28.8
		MCWB	24.8	25.5	26.4	27.1	27.5	27.9	27.5	27.2	27.0	26.5	26.1	25.0
	10%	DB	27.7	29.7	32.0	34.3	34.7	34.3	32.9	32.4	32.0	30.6	29.5	27.6
		MCWB	24.2	25.0	25.8	26.8	27.2	27.5	27.0	27.0	26.7	26.0	25.6	24.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.5	27.2	28.2	29.1	29.3	29.7	28.7	28.4	28.2	27.7	27.3	26.5
		MCDB	30.8	32.6	34.6	36.5	36.4	36.4	35.1	33.6	33.5	32.9	31.8	30.6
	2%	WB	25.7	26.6	27.4	28.2	28.5	28.6	28.1	27.8	27.6	27.1	26.7	25.8
		MCDB	29.7	31.7	33.6	35.3	35.8	35.2	34.2	33.1	32.8	31.8	31.0	29.3
	5%	WB	25.0	25.9	26.7	27.6	27.9	28.1	27.6	27.4	27.2	26.7	26.3	25.2
		MCDB	28.6	30.7	32.6	34.4	34.9	34.5	33.3	32.5	32.1	31.0	30.1	28.2
	10%	WB	24.3	25.2	26.1	27.0	27.3	27.6	27.1	27.0	26.8	26.2	25.7	24.7
		MCDB	27.4	29.3	31.4	33.3	33.7	33.6	32.4	31.9	31.3	30.0	29.1	27.2
Mean Daily Temperature Range	5% DB	MDBR	6.8	8.1	9.2	10.2	10.2	9.5	8.3	7.6	7.6	7.0	6.2	5.6
		MCDBR	8.7	10.5	11.3	11.7	11.8	10.8	10.1	9.0	9.0	8.8	8.5	7.6
		MCWBR	5.5	6.1	5.8	5.0	4.9	4.9	4.3	4.4	4.4	4.5	4.7	4.6
	5% WB	MCDBR	8.3	10.0	10.9	11.1	11.2	10.3	9.5	8.5	8.6	8.1	7.8	7.0
MCWBR		5.5	6.3	6.3	5.8	5.5	5.0	4.9	4.4	4.4	4.6	4.7	4.5	
Clear-Sky Solar Irradiance	taub	0.423	0.421	0.469	0.478	0.458	0.435	0.427	0.432	0.444	0.452	0.449	0.458	
	taud	2.361	2.374	2.236	2.231	2.318	2.390	2.413	2.391	2.358	2.318	2.319	2.257	
	Ebn at Noon	859	882	848	838	844	855	863	867	858	841	826	810	
	Edh at Noon	118	122	144	145	131	121	118	122	125	127	122	127	
All-Sky Solar Radiation	RadAvg	3.49	4.24	4.95	5.91	5.88	5.91	5.35	4.99	4.94	4.17	3.54	2.96	
	RadStd	0.32	0.60	0.48	0.44	0.45	0.45	0.47	0.35	0.34	0.32	0.49	0.35	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	-0.33	-0.24	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page